Fuzzy Emulated Symbolic Regression for Modelling and Control of Markov Jump Systems With Unknown Transition Rates

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***Abstract*—In this brief, a novel fuzzy emulated symbolic regression (FESR) model is proposed for modelling and control of Markov jump systems with unknown transition rates. Conventional symbolic regression model is improved in multi-layered form where internal functions, operations and sparse connections are determined via random-learning strategy. A clustering based fuzzy system is designed as a preprocessing-layer that brings an additional power to the model capability. Proposed model includes small number of the output parameters so it is parametrically parsimonious, implementable and easy designed model for future embedded-design applications. In numerical applications first, open-loop black-box modelling results of nonlinear Markov jump systems are shown to discuss the model accuracies where the systems are excited by multi-frequency sine inputs. Modelling results are compared in terms of mean squared-error (MSE) and minimum-descriptive length (MDL) criteria. Second, generalized predictive controller is designed to control the Markov jump systems with proposed model and unknown transition rate where stabilization results are discussed for future applications.**

***Index Terms*—Symbolic regression, parsimonious modelling, fuzzy preprocessing, predictive control, Markov jump system.**

I. Introduction

MACHINE learning models have been recently addressed to solve complex pattern recognition problems with high accuracy but they suffer from large-number of parameters. From control theory point of view, it is difficult to apply these large-sized architectures in embedded system design since they need large-memory size and more computations to produce a suitable control signal. Therefore, efficient, parsiminous and easy designed models are still promising for modelling and control of nonlinear systems.

Symbolic regression is an analytical representation of input-output functionals where genetic programming is used to optimize such predefined function-operation candidates to minimize the cost function of the regression [1]. Due to its implementable structure, symbolic regression has been recently applied for identification and control of nonlinear systems [2], [3]. Informative data samples have been selected for modeling of autonomous robots using symbolic regression [4]. As a shallow model, extreme learning machines were introduced in the advantage of high accuracy, easy design and fast-learning capability [5]. They are fundamentally one-hidden layer artificial neural-networks where the initial layer parameters are randomly assigned and output weighting parameters are usually optimized via least-squares estimation (LSE) [6], [7]. Recently, extreme learning machine based symbolic regression models have been introduced for modelling of nonlinear systems [8], [9] such that the power of the various symbolic functions and operators are combined with random learning to select hidden features from the input data.

Markov jump systems (MJSs) are a class of the stochastic systems with random transition matrices where some parameters change abruptly in a probability. Therefore, there is required robust modeling and control methods since it is difficult to produce suitable control signals during the change of parameters or Markov states. Recently, nonfragile fuzzy controller with optimal upper-bound of the guaranteed cost is proposed for time-delayed fuzzy MJSs in [10], and an observer-based stabilization [11] is proposed for exponential stability of the MJSs. Finite-time state-feedback controller has been designed for MJSs based on fuzzy hidden Markov modeling [12], fuzzy system based fault detection filter is proposed with asynchronous transmission times [13] and finite-region H_∞ asynchronous control has been designed for 2-D MJSs [14]. If the transition rate is unknown, model-based controllers can not efficiently control MJSs [15]. Recently, multivehicle tracking systems are modeled as Markov processes and controlled with unknown transition rates in [16], [17].

Following that the main contribution of this work is to propose an efficient and parsimonious regression model for modelling and control of nonlinear Markov jump systems with unknown transition rates. The advantages of symbolic regression, multi-layered form, fuzzy feature extraction, random-learning theory and sparse connections of internal layers are used to construct the proposed model namely "fuzzy emulated symbolic regression (FESR) model". The power of the random-learning strategy is utilized for selecting informative symbolic functions, operators and sparse connections. Markov jump systems with unknown mathematical model and unknown transition rates are difficult for modeling and control. In the open-loop identification, the proposed FESR model learns accurately the behavior of the

Manuscript received August 3, 2021; accepted August 11, 2021. Date of publication August 16, 2021; date of current version March 15, 2022. This brief was recommended by Associate Editor I. W. H. Ho.

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TCSII.2021.3104874.

Digital Object Identifier 10.1109/TCSII.2021.3104874

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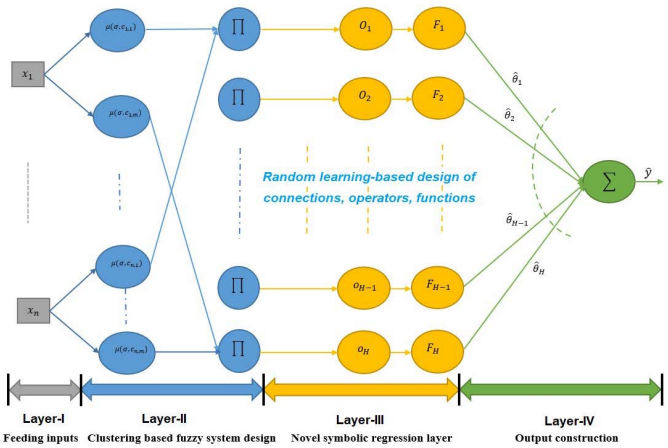

Fig. 1. Proposed FESR model.

MJS with different transitions. After that suitable predictive control signal is produced with FESR model without performance degrade. In numerical applications, comparative results are given in terms of the mean squared-error and minimum-descriptive length where better performances are provided by the proposed model.

II. Fuzzy Emulated Symbolic Regression

The proposed FESR model is proposed based on three building blocks as fuzzy preprocessing in the first-layer, symbolic regression with sparse connections in the second-layer and random-learning based design of second-layer. In fact, they are main differences and contributions compared to the tree-based genetic programming [1]. The proposed FESR model is shown in Figure 1.

Layer-I Input Layer: Input data of the model can be used in their domain due to the clustering in the second layer.

Layer-II Fuzzy Preprocessing: Clustering is one of the unsupervised learning method such that based on measures between data pairs, they are separated or clustered into clusters. Fuzzy C-means clustering both generates cluster centers and fuzzy memberships corresponding to the clusters. The memberships represent what degree a data belonging to the each cluster. Therefore, these cluster centers are suitable for the construction of the fuzzy system modelling. When Gaussian membership function is used, then it needs true centers to represent input data. The standard deviation can be taken as a constant for the fuzzy membership functions. Fuzzy C-means clustering [18] evaluates the membership values and centers as

$$\mu_{ik} = \frac{1}{\sum_{j=1}^{C}\left(\frac{||x_k - c_i||}{||x_k - c_j||}\right)^{2/(m-1)}} \qquad 1 \le i \le C$$

$$c_i = \frac{\sum_{i=1}^{N}(\mu_{ik})^m x_k}{\sum_{i=1}^{N}(\mu_{ik})^m} \qquad 1 \le k \le N \tag{1}$$

to minimize a cost function

$$J_m = \sum_{i=1}^{N}\sum_{j=1}^{C}(\mu_{ik})^m ||x_k - c_j||^2 \qquad 1 < m < \infty \tag{2}$$

where $x_k, 1 <= k <= N$ is the input data, $\mu(.)$ are membership functions, $c_i, \quad 1 <= i <= C$ are the centers, $J_m, \quad 1 < m < \infty$ cost values corresponding to the m batch iteration. The clustering is terminated when $||U^{j+1} - U^j|| < \epsilon$, where the U matrix is built up with membership values of data points. ϵ is a small constant to stop clustering algorithm where j is the corresponding center.

TABLE I
Some Operators and Functions

Operator	Function
$+$	x^2
$-$	$\sqrt{x}$
$\times$	$\cos(x)$
min	$\tanh(x)$
max	$1/(1+e^{-x})$
mod	$\max\{0, x\}$

Clustering based fuzzy system is designed in Mamdani-type rule base where j*th* rule is defined as

$$R^j : IF\ x_i\ is\ A_1^j\ and\ \cdots\ and\ x_i\ is\ A_M^j\ THEN\ y\ is\ B^j. \tag{3}$$

The output fuzzy basis functions are constructed using product inference engine, singleton fuzzifier and center average defuzzifier as

$$\boldsymbol{\phi}_j(x_i) = \frac{\prod_{i=1}^{N}\mu_{A_i^j}(x_i)}{\sum_{j=1}^{C}\prod_{i=1}^{N}\mu_{A_i^j}(x_i)}. \tag{4}$$

Cluster centers are utilized for Gaussian membership functions of inputs with unity standard deviation given as

$$\mu_{A_i^j}(x_i) = e^{-\frac{1}{2}\left(\frac{(x_i - c_j)^2}{\sigma_j^2}\right)} \tag{5}$$

for the i*th* input data and j*th* cluster center. The clustering of the input data provides a representation of hidden features in a rule base, then fuzzy inference transfers the input-data to the nonlinear feature space with normalization. Therefore, clustering based fuzzy system enables to reach the hidden features. Each fuzzy basis function passes through symbolic function layer with sparse connections and constructs regressor matrix.

Layer-III Multilayered Symbolic Regression: Following to fuzzy system layer that the internal patterns of predefined operators and functions are randomly selected to construct the proposed model. The symbolic models are constructed based on the analytical functions and operators so that complex data pairs can be represented as symbolic models in the sequential and nested form. For instance, $y \in R$ is the output variable and the input variable $x = (x_1, x_2, \ldots, x_n) \in R^n$ can be expressed as

$$y = f(x) + \epsilon \tag{6}$$

where ϵ is the modeling error, $f(\cdot)$ is the regressor function which is the combination of mathematical operators (such as $+, \times, \ldots$ and arithmetic operations like $\exp(\cdot), \sin(\cdot), \ldots$). Some of the well-known analytical operators and functions are given in Table I. Selected operation set does not include division due to the zero division problem. The deepness of the symbolic regression is decreased via multi-layer parallel processing structure similar to the artificial neural-networks. To decrease the computational load of the multi-layer processing

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units, there are used sparse connections between the sequential layers. Thus, symbolic functions, operators and connections are chosen randomly to construct the output regressor matrix $\boldsymbol{F} = [F_1, \ldots, F_H]$ where H is the number of the symbolic functions on the output layer.

Layer-IV Parameter Optimization: Output weighting parameters are designed by LSE optimization similar to the improved fuzzy function model [18] and extreme learning machine [6]. Parameters are estimated as $\hat{\boldsymbol{\theta}} = \boldsymbol{F} \dagger y$ where $\boldsymbol{F}\dagger$ is Moore-Penrose inverse of regressor matrix $\boldsymbol{F}$.

Proposed model can be used to approximate any input-output behavior when there is used sufficient number of symbolic functions similar to the polynomial and symbolic regression. Owing to unknown disturbances or system behavior, the output parameters continue to the online adaptation in control period without changing cluster centers, sparse connections, symbolic functions and operators. The design steps of the proposed model are given as

- Cluster the input data,
- Fuzzy system design based on clusters,
- Choose random connections of internal connections,
- Choose random operators for processing units,
- Choose random functions for processing units,
- Optimize the output weighting parameters.

III. Generalized Predictive Control

Model-predictive control is a sophisticated controller due to optimization-based predictive generation of control signal. In case of unknown system model, various function approximators can be used to predict the future behavior of the system [19]. For the nth order nonlinear discretized system, nonlinear generalized-predictive controller is designed as

$$\begin{aligned}
&\min_{\boldsymbol{u}} F(x_k, u_k)\\
&\text{subject to}\\
&\mathrm{X}_i = \left\{x_i \in \Re \mid \quad x_{i_{min}} \le x_i \le x_{i_{max}}, i = 1, \ldots, n_x\right\}\\
&\mathrm{U}_r = \left\{u_r \in \Re \mid \quad u_{r_{min}} \le u_r \le u_{r_{max}}, r = 1, \ldots, n_u\right\}\\
&\Delta\mathrm{U}_r = \left\{\Delta u_r \in \Re \mid \quad |\Delta u_r| \le \Delta u_{r_{max}}, r = 1, \ldots, n_{ud}\right\}\\
&\mathrm{Y}_q = \left\{y_q \in \Re \mid \quad y_{q_{min}} \le y_q \le y_{q_{max}}, q = 1, \ldots, n_y\right\}
\end{aligned} \tag{7}$$

with

$$F(u_k) = \frac{1}{2}\sum_{p=1}^{\kappa}\sum_{j=1}^{n_y}\left(\tilde{y}_{j|k+p} - \hat{y}_{j|k+p}\right)^2 + \frac{1}{2}\lambda_r\sum_{r=1}^{n_{ud}}\left(u_{r|k+1} - u_{r|k}\right)^2, \tag{8}$$

where λ_r is the penalty term that prevents undesired changes on the input signal, and κ is the prediction horizon of the output. $\hat{y}_{q|k+\kappa}$ is the estimated output for the qth sample and $\tilde{\boldsymbol{y}}$ is the reference signal. It is assumed that the short-term future values are known for the reference signal. Furthermore, there are some characteristics in (8) such as input/input slew constraints which occur due to actuator limitations and state/output constraints which are related to operational limitations. When $\hat{\boldsymbol{e}}$ is the prediction error vector as

$$\hat{\boldsymbol{e}} = \begin{bmatrix} \tilde{y}_{1|k+1} - \hat{y}_{1|k+1} \\ \vdots \\ \tilde{y}_{q|k+\kappa} - \hat{y}_{q|k+\kappa} \\ \sqrt{\lambda_1}\Delta\boldsymbol{u}_{k+1} \\ \vdots \\ \sqrt{\lambda_{\mathrm{R}}}\Delta\boldsymbol{u}_{k+1} \end{bmatrix} \tag{9}$$

$\boldsymbol{J}_u$ matrix is defined as

$$\boldsymbol{J}_u = -\begin{bmatrix} \frac{\partial\hat{y}_{1|k+1}}{\partial u_{1|k+1}} & \frac{\partial\hat{y}_{1|k+1}}{\partial u_{2|k+1}} & \cdots & \frac{\partial\hat{y}_{1|k+1}}{\partial u_{r|k+1}} \\ & \vdots & & \\ \frac{\partial\hat{y}_{q|k+\kappa}}{\partial u_{1|k+1}} & \frac{\partial\hat{y}_{q|k+\kappa}}{\partial u_{2|k+1}} & \cdots & \frac{\partial\hat{y}_{q|k+\kappa}}{\partial u_{r|k+1}} \\ \sqrt{\lambda_1} & \sqrt{\lambda_1} & \cdots & \sqrt{\lambda_1} \\ \sqrt{\lambda_{\mathrm{R}}} & \sqrt{\lambda_{\mathrm{R}}} & \cdots & \sqrt{\lambda_{\mathrm{R}}} \end{bmatrix}. \tag{10}$$

The update rule for $\boldsymbol{u}_{k+1}$ is written as

$$\boldsymbol{u}_{k+1} \Longleftarrow \boldsymbol{u}_k + \delta\boldsymbol{u}_{k+1}, \tag{11}$$

$$\delta\boldsymbol{u}_{k+1} = -\left(\boldsymbol{J}_u^T\boldsymbol{J}_u + \mu_u\boldsymbol{I}\right)^{-1}\boldsymbol{J}_u^T\hat{\boldsymbol{e}} \tag{12}$$

The variation of the control signal can be optimally found using (12). However, the mathematical model of the system or approximate model of the system is needed for the Jacobian calculation. In [19], the system model is discretized using Runge-Kutta (RK) 4th order integration, based on the integration model, exact Jacobian is calculated and control signal is produced. The state and output at $(n+1)th$ sampling index can be written as

$$\begin{aligned}
\boldsymbol{x}(n+1) &= \boldsymbol{x}(n) + \frac{1}{6}K_1\boldsymbol{F} + \frac{1}{3}K_2\boldsymbol{F} + \frac{1}{3}K_3\boldsymbol{F} + \frac{1}{6}K_4\boldsymbol{F},\\
y(n+1) &= g(\boldsymbol{x}(n+1), \boldsymbol{u}(n)).
\end{aligned} \tag{13}$$

The K_iF_j variables with $i = 1, \ldots, 4$ and $j = 1, \ldots, N$ are defined as

$$\begin{aligned}
K_1\boldsymbol{F} &= T_s\boldsymbol{f}(\boldsymbol{x}(n), \boldsymbol{u}(n)),\\
K_2\boldsymbol{F} &= T_s\boldsymbol{f}\left(\boldsymbol{x}(n) + \frac{1}{2}K_1\boldsymbol{F}, \boldsymbol{u}(n)\right)\\
K_3\boldsymbol{F} &= T_s\boldsymbol{f}\left(\boldsymbol{x}(n) + \frac{1}{2}K_2\boldsymbol{F}, \boldsymbol{u}(n)\right)\\
K_4\boldsymbol{F} &= T_s\boldsymbol{f}(\boldsymbol{x}(n) + K_3\boldsymbol{F}, \boldsymbol{u}(n)),
\end{aligned} \tag{14}$$

where it is referred to as the RK discretized model of the system. For a Lipschitz nonlinear system, stability of discretization can be shown using sufficiently small step-size. At the same time, the unknown model parameters of the system can be updated based on the second-order gradient descent approach. The future behavior of the reference signal is directly embedded to the parameter change, then parameter adaptations are driven to minimize the cost function.

IV. Computational Results

A. Markov Jump Systems

Consider a continuous-time single-input single-output nonlinear system,

$$\begin{aligned}
\dot{\boldsymbol{x}} &= \boldsymbol{f}(\boldsymbol{x}, u, r_t)\\
y &= h(\boldsymbol{x})
\end{aligned} \tag{15}$$

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where $\boldsymbol{x} \in \Re^n$ is the state vector, $u \in \Re$ is the input, $y \in \Re$ is the output, r_t is the Markov Jump parameter. The $\boldsymbol{f}(.) \in \Re^n$ and $h(.) \in \Re$ are nonlinear differentiable functions with $\|\dot{f}(t)\| \leq f$ for $0 \leq f \leq \infty$. It is assumed that r_t is time-varying parameter based on discrete-time Markov stochastic process, taking values from a set of finite states $r \in S = 1, 2, \ldots, N$. The state transition matrix of the parameter is $\Pi = \pi_{rk}$, r, $k \in S$. The transition probability can be described as

$$P_r\{r_{t+\Delta t} = k \mid r_t = r\} = \begin{cases} \pi_{rk}\Delta t + o(\Delta t), & r \neq k \\ 1 + \pi_{rr}\Delta t + o(\Delta t), & r = k, \end{cases} \quad (16)$$

where $\Delta t > 0$ and $\lim_{\Delta t \downarrow 0} o(\Delta t)/\Delta t \to 0$. $\pi_{rk} \geq 0$ is the transition rates from mode r at time t and from k at time $t + \Delta t$ with $\pi_{rr} = -\sum_{k=1, k\neq r}^{N} \pi_{rk}$.

B. Single-Link Robot Arm

Single-link robot-arm presents a Markov process behavior due to the change of parameters. Its dynamics include a simple nonlinear-term but parameters change with a probability that makes it difficult to control therefore it has been used to test the proposed models and controllers [10]. The dynamics are given as

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -\frac{glM_k}{J_k}\sin(x_1(t)) - \frac{R}{J_k}x_2(t) + \frac{1}{J_k}u(t) \end{aligned} \quad (17)$$

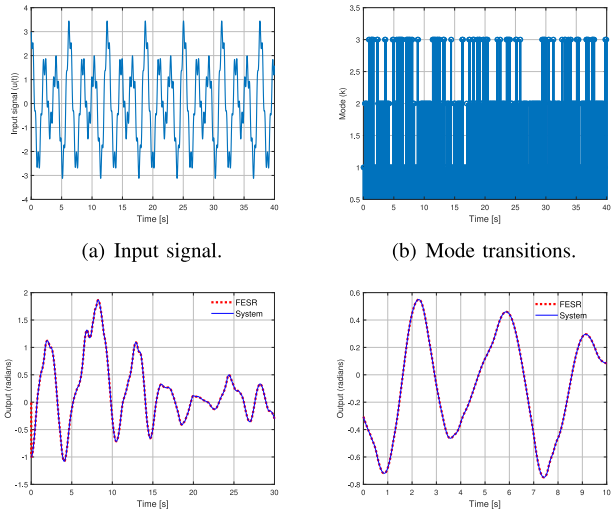
where $x_1(t)$ is the angular position and $x_2(t)$ is the angular velocity, respectively. The fixed system parameters are $R = 2, l = 0.5, g = 9.81\ m/s^2$. However, M_k and J_k vary as time goes that exhibit Markov process with three states as $M_1 = J_1 = 1$, $M_2 = J_2 = 4$ and $M_3 = J_3 = 10$, respectively. The transition probability matrix is given as

$$\boldsymbol{P} = \begin{bmatrix} 0.2 & 0.65 & 0.15 \\ 0.1 & 0.4 & 0.5 \\ 0.7 & 0.07 & 0.23 \end{bmatrix}. \quad (18)$$

In numerical applications, first physical dynamics of the robot-arm is modelled by the proposed FESR model in open-loop identification, then generalized predictive controller is designed to stabilize its position using designed identification model.

C. Open-Loop Identification

The input-output model of MJS is constructed in the form of $\hat{y}(k) = \hat{F}(u(k), y(k-1))$ where each input is clustered in $C = 5$ clusters. The constructed FESR model has 3 layers where output-layer has 50 functions and operators that are weighted to approximate the system output. For the modeling of robot-arm, the system is excited by $u(t) = 2\cos(0.01t) - 3\sin(0.05t) - 0.5\sin(0.01t)\cos(0.5t)$ where Euler discretization is used for integration routine with 0.01 seconds sampling-time. The sine signal with four different frequencies and random change of the system parameter can provide persistent excitation for identification task. The first 30 seconds of collected data is used for training, and remaining 10 seconds of collected data is utilized for testing, respectively. The input signal is applied in open-loop framework then input-output modeling results are given in the following figures.



(a) Input signal. (b) Mode transitions.

(c) Training results. (d) Testing results.

Fig. 2. Open-loop identification results of robot-arm.

TABLE II
PERFORMANCE MEASURES FOR MODELLING

	MSE of Test Data	MDL	C.Time [s]
ANFIS [20]	20.0e-3	-4.22e4	65.6
ELM-SR [8]	3.19e-4	-5.23e4	21.6
FESR (proposed)	2.79e-5	-6.82e4	27.2

Figure 2 presents the identification results of robot-arm using the proposed model. Figure 2(a) presents artificial input signal, Figure 2(b) gives the mode transitions of the stochastic parameter. Finally, Figure 2(c) and Figure 2(d) show the training and testing results of the proposed model, respectively.

The performances of designed regression models are compared in Table II. Conventional adaptive-neuro fuzzy-inference system (ANFIS) [20], symbolic regression based extreme-learning machine model (ELM-SR) [8] and proposed FESR model are compared for modeling of robot-arm. ANFIS model is designed based on MATLAB toolbox with 100 training epoches and 5 input membership functions same as the number of clusters. ELM-SR and FESR models are constructed with 30 random evoluations to be statistically reasonable for symbolic functions, operations and sparse connections. The mean squared-error value is calculated as $\text{MSE} = e^T e/(2N)$ where N is the number of data, e is the modelling error. The minimum descriptive-length criteria is measured by $\text{MDL} = N\log(E) + P\log(N)$ where N is the number of data, P is the number of parameters, E is the modelling error of training data. Small MDL criterion means that parameters of the model are more informative and so the modeling capability is better than others. In the last column of the table, computational time of the models are compared. According to the performance measures, MSE performance of testing data and MDL performance of the model are much improved using the proposed FESR model. ANFIS model presents also small MDL value due to the less number of parameters and small training MSE performance but its testing performance is not acceptable. Designed and compared regressor models need acceptable computational-times for re-modeling of the system in internal periods of control process.

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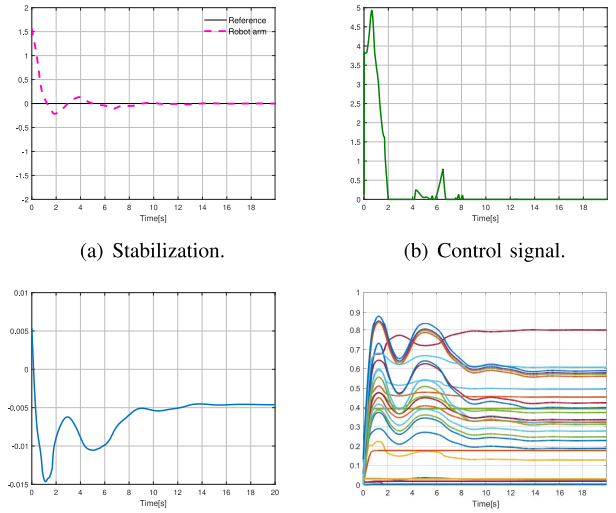

(a) Stabilization. (b) Control signal.

(c) Modelling error. (d) Adaptation of parameters.

Fig. 3. Stabilization results of robot-arm.

D. Predictive Control With Unknown Transitions

For the control of robot arm, stabilization is performed during unknown Markov transitions exist. Since the mathematical model of the system is assumed as unknown, the proposed predictive controller employs the designed FESR model for output predictions of the MJS. The output parameters of the designed FESR model are normalized close to zero then continued to the online gradient-based adaptation for generalized predictive control for unknown behavior of the system and disturbances. Remember that the main parts of the designed FESR model such as cluster centers, symbolic functions and operators, and sparse connections are not changed in the control.

For the robot arm, stabilization results are shown in Figure 3. Figure 3(a) presents the end-effector position where Figure 3(b) indicates the corresponding control signal. The online modelling errors and adaptation of model parameters under predictive control are presented in Figure 3(c) and Figure 3(d), respectively. There exist difficulty for the parameter selection of generalized predictive control such that $K_y = 5, K_u = 1$ and $\Delta u_{max} = 0.5$ are selected in the applications. It should be known that the control accuracy of the generalized predictive controller with unknown transition rates is here provided by the proposed FESR model. Due to the different transition rates and parameter designs, the application results are not compared with methods in literature works. In general framework, the results of stabilization are acceptable for a real-time implementation such that stabilization is provided in two seconds with small oscillations.

V. Conclusion

A novel regression model including symbolic regression, fuzzy preprocessing and random learning strategy was proposed and applied for modelling and control of Markov Jump systems. The idea of proposed model is to construct a parsimonious model with high-modelling accuracy. First, open-loop identification results of a robot-arm were given with highly changing transitions. Second, generalized predictive controller was designed and applied to control. Due to the less number of parameters, it can be easily realized in embedded systems. Multi-class classification and adaptive control are ongoing works using the developed model.

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